

Data Science workflow

- ① Domain Analysis
- ② Data Collection
- ③ Data Understanding
- ④ Data cleaning
- ⑤ EDA $\rightarrow$ uni, bi,
- ⑥ Feature Engg
- ⑦ Preprocessing
- ⑧ Train-test-split
- ⑨ Model Training
- ⑩ Model Evaluation
- ⑪ hyper parameter tuning
- ⑫ Deployment
- ⑬ Monitor

Index	LID	loan amount	Gender	Married
0	-	-	-	Yes
1				Yes
→ 2				Null
3				Yes
→ 4				Null
→ 5				Null
6				Yes

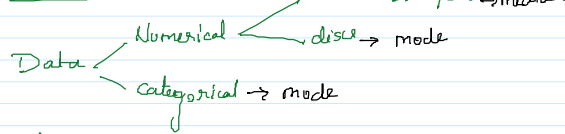
df.loc[row col]
 $\downarrow$
df.loc[df.married.isnull()]

1) Detect & Handle the Null values

Detect

- $\rightarrow$ info()
- $\rightarrow$ describe()
- $\rightarrow$ isnull()

Handle the Null values $\rightarrow$



Age
 35
 Null $\rightarrow$ replace?
 40
 Null
 65
 75

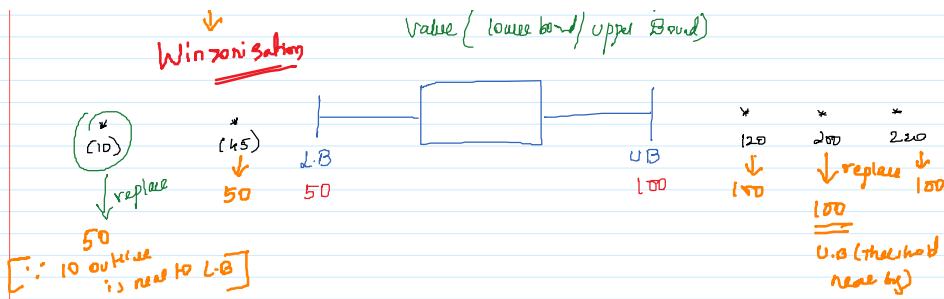
II Detect & Handle the Outliers

- a) Detect $\rightarrow$ methods
- ① boxplot
 - ② IQR $\rightarrow$ $\left. \begin{array}{l} \text{val} < \text{Low-bound} \\ \text{val} > \text{Upper-bound} \end{array} \right\} \rightarrow$ define Outliers
 - ③ Zscore
 - ④ Empirical rule $\rightarrow$ value is beyond the limit
 $\mu - 3\sigma$ to $\mu + 3\sigma$

b) Handle the outliers

- a) 2-3% drop/delete the Outliers
- b) Capping method $\rightarrow$ outlier values are replaced with nearby threshold value (lower bound/upper Bound)

Winorisation



Scaling is the process of transforming numerical data so that features are on a similar scale

Scaling :- Convert all Numerical data into one form

- ① MinMax scales → 0 to 1 → deep Algo, Atriv, Image Extract
- ② Standard scales → -3 to +3 → Distance based Algo
- ③ Robust scales → median, Iqr → Handles outlier, Outliers

Min-max

$$\text{Min-max} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Standard scales (Zscore)

$$Z_{\text{score}} = \frac{X - \text{Mean}}{\sigma}$$

Robust scales

$$R = \frac{X - \text{Median}}{\text{IQR}}$$

- ① Null values
- ② Duplication
- ③ Outliers
- ④ Transformation
- ⑤ Encoding
- ⑥ Scaling

Encoding :- Handling the categorical data

→ Categorical data is converted to Numerical data

Eg:- Female → 0
 Male → 1

Types of Categorical data

- Nominal data → doesn't have order
- Ordinal data → order

① OHE → One Hot Encoding Technique

↳ Nominal

Gender	Gender - Female	Gender - Male
Male	0	1
Male	0	1
Female	1	0
Male	0	1
Female	1	0

Colour	Colour - Red	Colour - Green	Colour - Blue
Red	1	0	0
Green	0	1	0
Blue	0	0	1

<u>Colour</u>	<u>Colour - Red</u>	<u>Colour - Green</u>	<u>Colour - Blue</u>
Red	1	0	0
Green	0	1	0
Red	1	0	0
Blue	0	0	1
Green	0	1	0
Blue	0	0	1

2) Label encoder

<u>Gender</u>	<u>Gender-label</u>
Female	0
Male	1
Female	0
Male	1

(Alphabetic)
 Unique → Female → 0
 Male → 1

Target Var — X
 ↓
 ML autom

Colour
 Red
 Green
 Blue

Alph → Blue → 0
 Green → 1
 Red → 2

3) Ordinal encoder → Ordinal data

Size → S, M, L, XL, XXL → define
 ↓ ↓ ↓ ↓ ↓
 0 1 2 3 4

Eg: Principal → 2
 Student → 0
 Teacher → 1

4) frequency encoder

<u>Colour</u>	
Red	→ 3
Green	→ 4
Red	→ 3
Blue	→ 2
Green	→ 4
Red	→ 3
Blue	→ 2
Green	→ 4
Green	→ 4

freq
 Red → 3
 Green → 4
 Blue → 2

5) Target encoder

place
 Bombay → 3
 Bang → 1
 Delhi → 2
 Delhi → 2
 Bombay → 3
 Bombay → 3

Encoding Categorical data
 Feature → Handle the categorical
 Target → Not handling categorical

Nominal - OHE, label enc X

Feature → model

Species
 Setosa → 0
 Vers → 1
 Vers → 2 } label encode automati

Feature

Colour → label encoder
 Red → 2
 Green → 1
 blue → 0

Feature
 OHE ✓
 $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ No relation

model $\frac{\text{label}}{2, 1, 0} \rightarrow \text{relation} \rightarrow \text{ranking}$

Summarize! Data Transformation $\begin{cases} \text{Normalisation} \\ \text{Encoding} \\ \text{Scaling} \end{cases}$

Transform → Skewed data → NP
 $\begin{cases} > 0 \rightarrow R \\ < 0 \rightarrow L \end{cases}$

moderate No Transf. More Skewness
 $\begin{cases} R \\ L \\ R+L \end{cases}$

Log x
 cube
 power
 log x+1

① +ve, -ve
 ② zero?
 ③ More skewed

Scaling $\begin{cases} \text{Feature} \checkmark \rightarrow \text{Continuous data} \\ \text{Target} \end{cases}$

iris → $\frac{SL \quad PL \quad PW \quad PL}{\text{same range of values}} \rightarrow \text{Scaling}$

Age | salary
 25 | 50000 → Scaling

* distance-based algo → Scaling is must

→ Standard

Neural Net
 L2 or L1
 min max

* Robust Scales

→ mainly when outliers present

Categorical $\begin{cases} \text{Encoding} \\ \text{OHE} \rightarrow \text{1D} \rightarrow \text{Vectors} \\ \text{label} \rightarrow \text{Alphabet} \rightarrow 0 \text{ to } n \\ \text{Ordinal} \rightarrow \text{Based on priority / ranking} \end{cases}$

✓ Freq → Based on freq of that value

④ KNN → k Nearest Neighbour

→ Supervised M.L

→ labelled data

→ principle : 1) Distance based algo
 2) prob based algorithm

→ Lazy algo

→ It works only / Active only when there is test data

→ No training the data

→ K → No. of Neighbours

→ K=3

→ Regression + classification

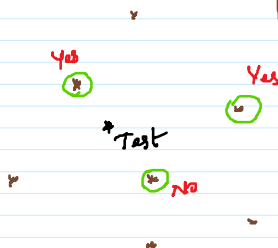


→ Training data
→ data
K=3



$$Reg = \frac{5+4+6}{3} = \text{avg of the values}$$

Classifi = K → odd No. 3, 5, 7
K=3



K=4
+4
+4
+4
+4

highest prob of class ⇒ Yes

Yes = 2

No = 1

predict from model ⇒ Yes

SAMPLING TECH

→ To make Dataset Balance

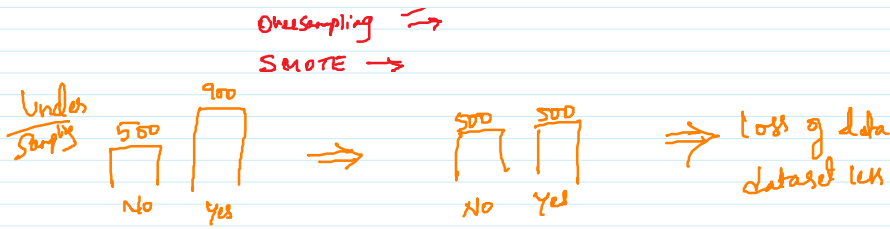
→ Classification

Yes → 900 records
No → 500 Records } Imbalanced

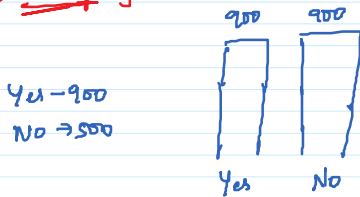
Yes → 900
No → 900 } Balanced

→ Imbalanced → Model is biased towards more records

Tech → Undersampling →



OverSampling



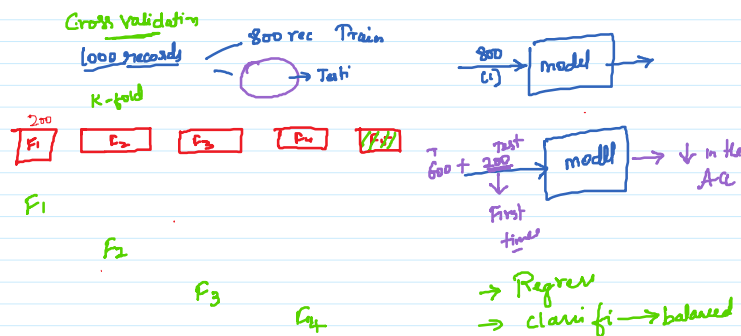
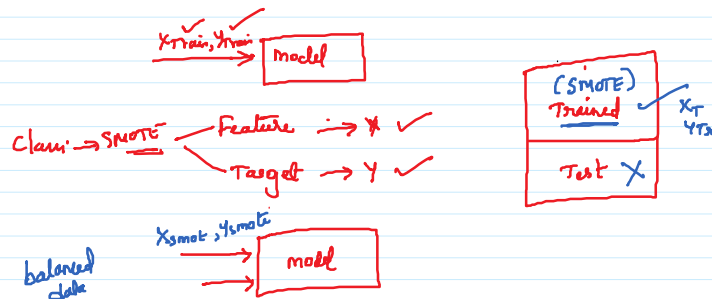
$\rightarrow$ duplicate values (1800)
 $\rightarrow$ Model $\rightarrow$ overfitting
Same data is trained multiple

Model 1
 $\rightarrow$ Training data $\Rightarrow$ 95% $\checkmark$
 $\rightarrow$ Test $\Rightarrow$ 60% $\times$
Overfitting

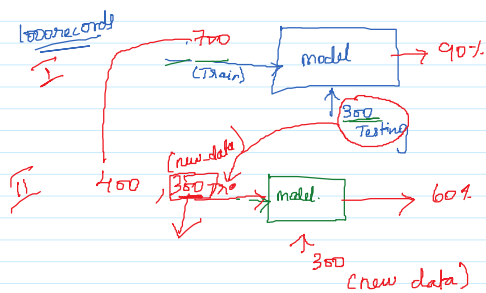
Model 2
Training Acc = 82% $\checkmark$
Test Acc = 80% $\checkmark$
Generalised Model

Model 3
Train = 45% $\times$
Test = 3% $\times$
Underfitting

SMOTE $\rightarrow$ Synthetic Minority OverSampling Technique



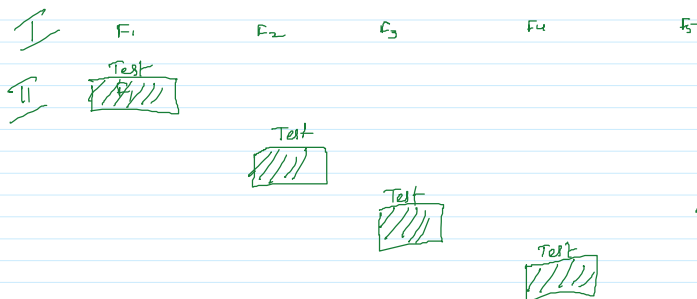
(4) How many times



- (1) How many times training model?
One time
- (2) Train Multiple

Cross Validation

80% Train 20% Testing



- (1) Regression
- (2) Classification
↳ Balanced

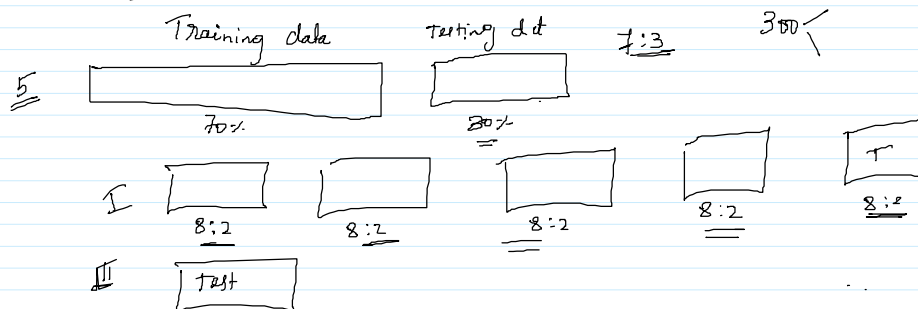


- (1) Regression
- (2) Classification
↳ Balanced

Stratified folding.

80:20

→ Unbalanced



PCA → Principal Component Analysis (Technique)

→ Unsupervised M.L

→ Reducing the dimensionalities // no. of columns

→ Unsupervised M.L

→ Reducing the dimensionalities (no. of columns)

Eg:- 100 col → 30 col
DRI PCA → 30

→ More time to compute

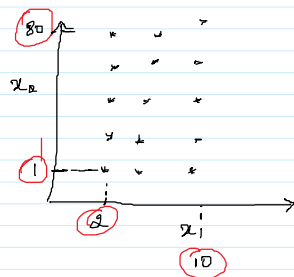
→ duplicate

→ More memory

→ Understand / EDA → More time

→ correlation

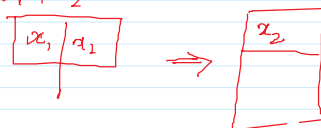
→ Principle: Co-Variance [Relationship b/w the feature]



x_1 is more Variance?

x_2 is more Vari

x_1 & x_2 are same



* Feature Extraction

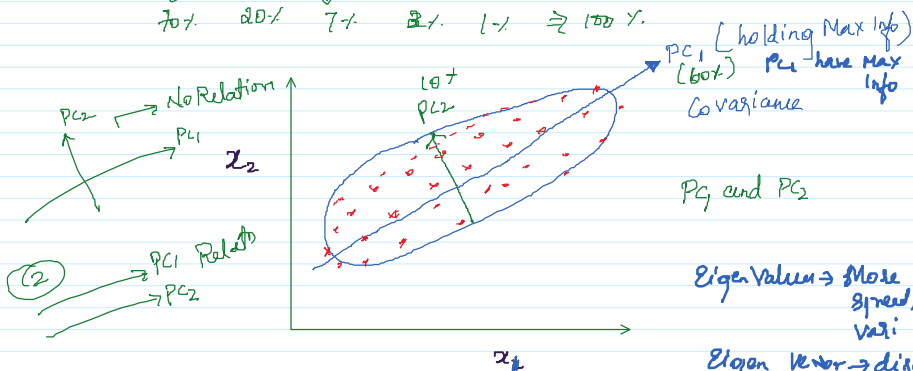
D.O.B	age
07-01-2004	16

* Feature Create

a	b	a+b

* How PCA Works?

F_1 F_2 F_3 F_4 F_5
↓ ↓ ↓ ↓ ↓
 PC_1 PC_2 PC_3 PC_4 PC_5
↓ ↓ ↓ ↓ ↓
70% 20% 7% 2% 1% → 100%



PC_1 and PC_2

Eigen Value → More spread Vari

Eigen Vector → direction

→ PC_1 always have Maximum Variance Ratio

EVR := Explained Variance Ratio

Assume $PC_1 = 40$

$PC_2 = 20$

$PC_3 = 5$

$EVR = \frac{\text{Variance } PC}{\text{total Var}}$

(corr)

$$\begin{bmatrix} 0.71 & 0.91 & 0.73 \\ & & \end{bmatrix}$$
 3x3
 correlation

$PC_1 = 40$
 $PC_2 = 20$
 $PC_3 = 5$

$EVR = \frac{\text{variance of PC}}{\text{total var}}$

$PC_1 EVR = \frac{40}{65} = \frac{40}{40+20+5} = 61.53$
 $PC_2 EVR = \frac{20}{65} = 30.7\%$
 $PC_3 EVR = \frac{5}{65} = 0.076$

Correlation
 Covariance
 $\text{cov}() (PC_1)$

3×3

100%

As

	F_1	F_2	F_3
Variance	40	20	5
EVR	61.53%	30.7%	0.076%

$3 \text{ Col} \rightarrow PC_1 + PC_2 = 90\%$
 Final Col
 $3 \text{ Col} \rightarrow 2 \text{ Col}$

x_1	x_2	x_3
54.91	71.61	81.31
42.61	81.22	16.33

PCA

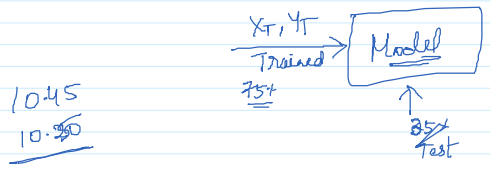
PC_1	PC_2
141	2.12
81.62	81.13

- Standardization $\rightarrow$ scaling the features ✓
- Covariance
- Eigen value & Eigen Vector
- find EVR
- Select the No. of Principal Component based on Max Info

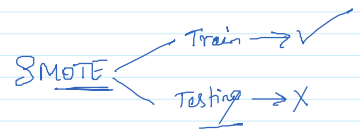
$\frac{dp}{dp} [70.05, 20.00, 5.00, \dots]$
 $\downarrow$
 PC_1 $\downarrow$ PC_2 $\downarrow$ PC_3

PC_1
 $PC_1 + PC_2$
 $PC_1 + PC_2 + PC_3$
 $PC_1 + PC_2 + PC_3 + \dots$

$\sim > 80\%$



balanced q/p $\leftarrow$ biased



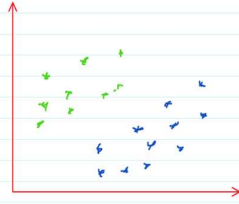
$\rightarrow X_T, Y_T, X_{\text{test}}, Y_{\text{test}}$
 $\rightarrow \text{SMOTE}(X_T, Y_T)$
 $\rightarrow CV = 5$
 + pred the model Test data?
 X_{test}
 $Y_{\text{test}}, Y_{\text{pred}}$

Y_{Test} Y_{pred}

7 April 2026 20:38

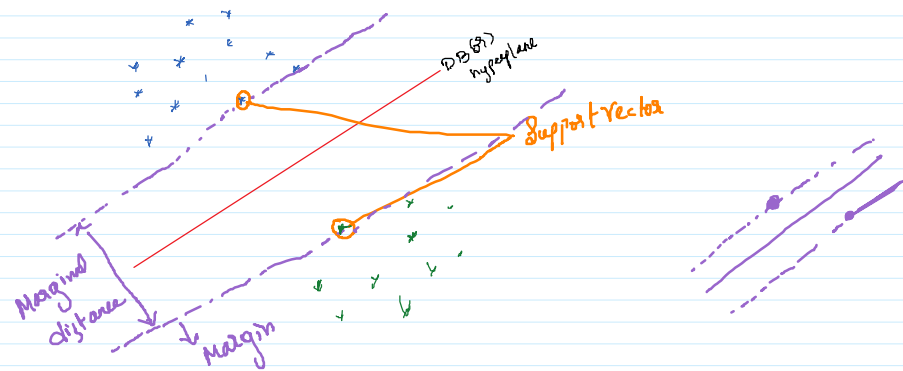
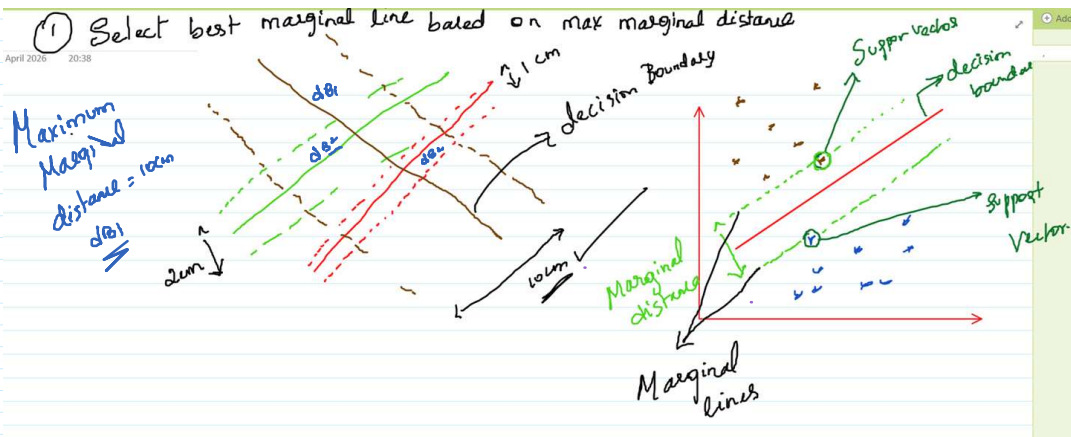
SVM → Support Vector Machine

- ① Supervised M.L algo
 - ② labelled data $\begin{cases} \text{IP} \rightarrow \text{Continuous} \rightarrow \text{Reg} \\ \text{OP} \rightarrow \text{discrete data} \rightarrow \text{Classifi} \end{cases}$
 - ③ SVM work on both Reg or classification
- SVM $\begin{cases} \text{SVR} \rightarrow \text{Support Vector Regression} \\ \text{SVC} \rightarrow \text{Support Vector Classification} \end{cases}$

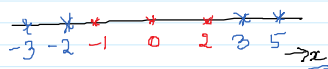


① Select best marginal line based on max marginal distance

April 2026 20:38



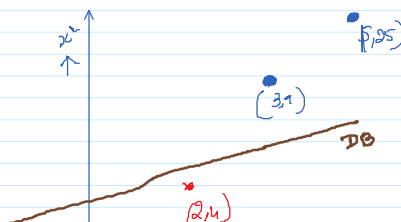
Non Linear Data



①
x 1 2

(3,7)

(2,4)

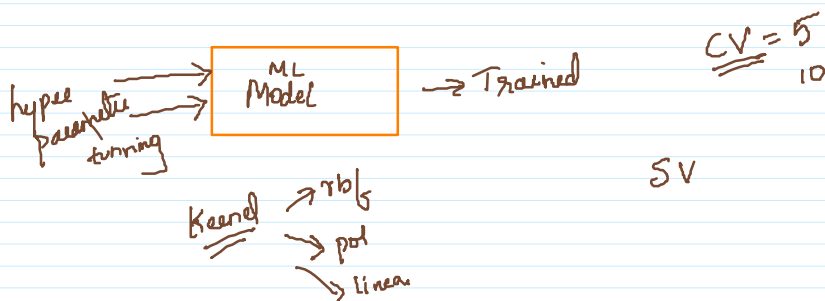
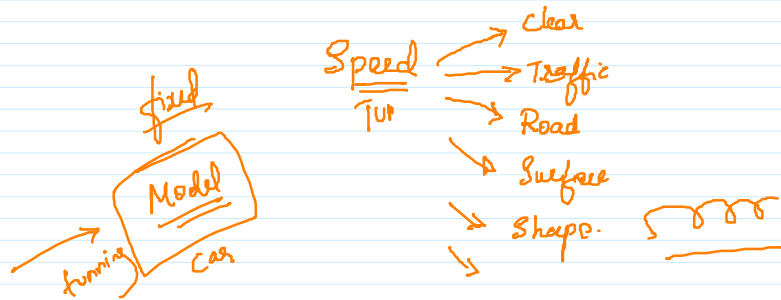


(1)

x	x^2
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9



Kernel $\rightarrow$ converts lower dimensionality to higher dimension
conv $\rightarrow$ linearly separable data



SVM $\rightarrow$ hyperparameter tuning (θ, w) $C = \text{less} \times$ $C = \text{more} \times$

① $C \rightarrow$ Regularization $\rightarrow$ how strictly our model should?

$\rightarrow$ underfit missclassification $\rightarrow$ low $\rightarrow C$ $0.1 \quad 0.01$

$\rightarrow$ Very Strict $\rightarrow$ Overfit $\rightarrow$ high $\rightarrow C$ $0.2 \quad 10 \quad 2$

$\rightarrow$ Marginal dist $\rightarrow C$ less $\rightarrow$ more marginal dist $\rightarrow$ more mistake

more $\rightarrow$ less marginal dist $\rightarrow$ error +

2) gamma $\rightarrow$ Shape of decision Bound
 $\rightarrow$ Non-linearly Separable data

$\rightarrow$ Kernel + RBF, poly, Sigmoid

$\gamma = \text{less}$  $\rightarrow$ underfitting
 $\gamma = \text{more}$  $\rightarrow$ overfitting

Svm $\rightarrow$ punish $C \Rightarrow$ less

Max Marginal (or) mis Ann.

Optimal value of γ & C

		T, F	
T, F		T	F
T	F	T	T
F	T	T	F
F	F	F	T
F	F	F	F

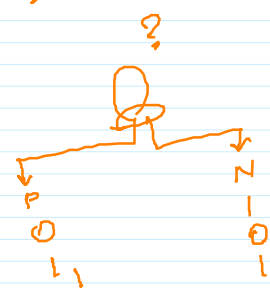
} all the combi

3 ip

0 0 1
0 1 0
0 1 1

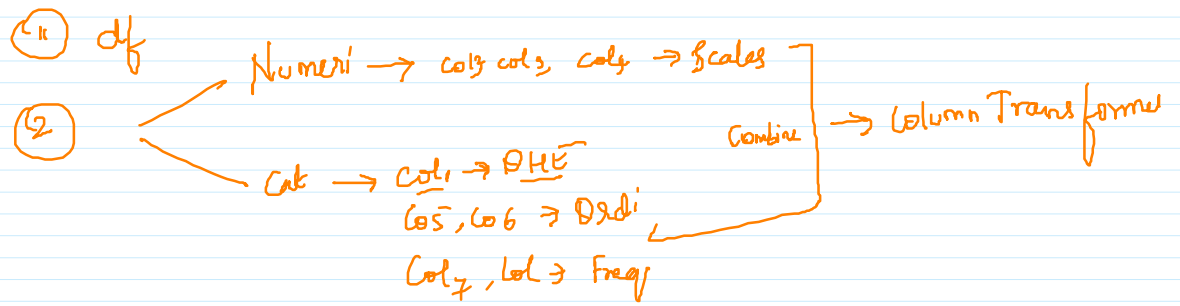
9 comb

10⁹ ip

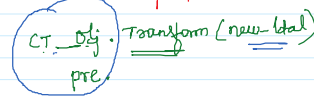
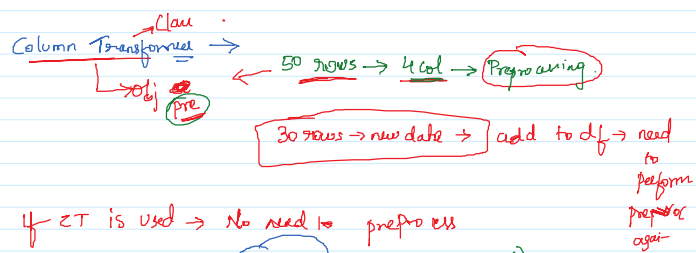
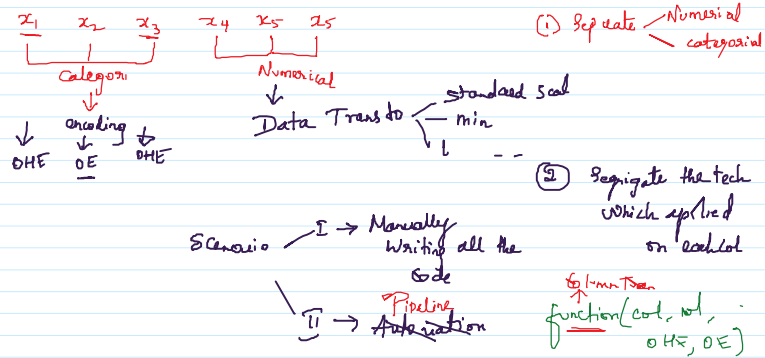


Grid Search CV $\rightarrow$ generate combinations

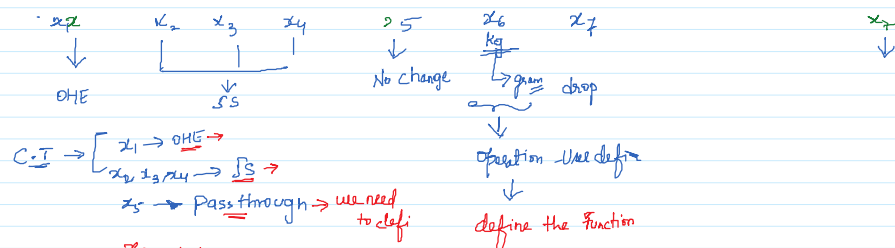
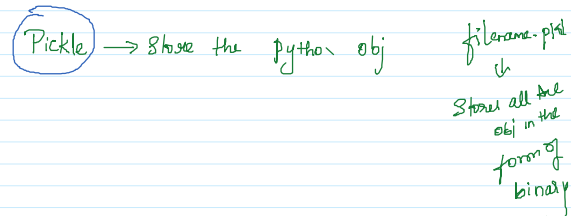
$\rightarrow$ Train the model.



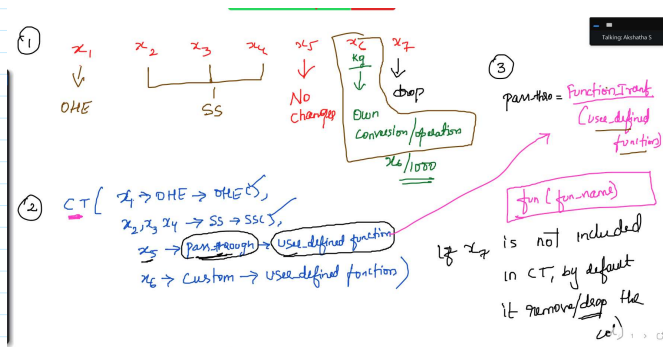
30-June-WDE



CT $\rightarrow$ class $\rightarrow$ obj $\rightarrow$ Pre-processing steps $\rightarrow$ stored in obj



$x_6 \rightarrow 0.12$



$x_5 \rightarrow \text{param_through} \rightarrow \text{FunctionTransformer (use-defined function)}$

$x_6 \rightarrow \text{Custom} \rightarrow 1$

years
months

def kg-gram(x):
return x/1000

use-defined function

Custom = FunctionTransformer(kg-gram)

1) $C = 0.01$ $\sqrt{q} = 1$ Ext $\rightarrow$ Sum $\rightarrow$ pred $\rightarrow 63.7\%$

$C = 0.01$ $\sqrt{q} = 10$ $\rightarrow$ Sum $\rightarrow -$

$C = 1$ $\sqrt{q} = 1.2$ $\rightarrow$ Sum

$C = 1$ $\sqrt{q} = 1.3$ $\rightarrow$ CV

Weather	Play	
Sunny	No	1
Overcast	Yes	2
Rainy	Yes	3
Sunny	Yes	4
Sunny	Yes	5
Overcast	Yes	6
Rainy	No	7
Rainy	No	8
Sunny	Yes	9
Rainy	Yes	10
Sunny	No	11
Overcast	Yes	12
Overcast	Yes	13
Rainy	No	14

Naive Bayes theorem

$\rightarrow$ probability based theorem

$\rightarrow$ Text Conversion

my name is Aksh $\rightarrow$ Number

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$P(\text{Yes}) = 9/14$$

$$P(\text{No}) = 5/14$$

$$P(\text{Sunny}) = 5/14$$

$$P(\text{Rainy}) = 5/14$$

$$P(\text{Overcast}) = 4/14$$

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

freq table

Weather	Yes	No
Sunny	3Y	2N
Overcast	4Y	0N
Rainy	2Y	3N
	9Y	5N

$$P(\text{Yes}) = \frac{9}{14}$$

$$P(\text{No}) = \frac{5}{14}$$

$$P(\text{Sunny}) = \frac{5}{14}$$

$$P(\text{Over}) = 4/14$$

$$P(\text{Rainy}) = 5/14$$

$$P\left(\frac{\text{Sunny}}{\text{Yes}}\right)$$

$$\text{Sunny? } ① P(\text{Yes/sunny}) = \frac{P(\text{sunny/yes}) \times P(\text{yes})}{P(\text{yes})} \Rightarrow \frac{3/9 \times 9/14}{5/14}$$

$$P(\text{No/sunny}) = 40\%$$

$$= 60\%$$